

Thomas John Neuenfeldt

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EDUCATION

University of Michigan

Ann Arbor, MI

Bachelor of Science Engineering in Computer Science; Minor in Music | GPA: 3.96 / 4.0

December 2025

Coursework: Linear Algebra, Discrete Mathematics, Data Structures and Algorithms, Machine Learning, Computer Vision, Human Robot Interaction, Robot Localization and Mapping, Stats & Data Analysis, Operating Systems, Database Management

SKILLS

Programming Languages: Java, C/C++, Python, Django, Julia, MATLAB/Simulink, SQL, HTML, JavaScript, CSS

Libraries: PyTorch, OpenCV, MongoDB, OpenSearch, Elasticsearch

Developer Tools: Django, Vue.js, ROS, Linux/Unix systems, Git

WORK EXPERIENCE

Amazon Web Services

Palo Alto, CA

Software Development Engineer Intern | Java, AWS, Agile Methodology

May 2025 – July 2025

- Designed and implemented enhancements to AWS' OpenSearch profiler based on stakeholder requirements, allowing customers to reduce latency for millions of queries made daily and lower their operational cost.
- Identified and resolved bugs proactively in OpenSearch to improve functionality and performance. Collaborated in weekly Sprint meetings to drive project progress. Collected external stakeholder requirements to deliver customer-obsessed solutions.

GE Vernova

Schenectady, NY

Onshore Wind Fleet Operations Intern

May 2024 – August 2024

- Optimized proprietary wind turbine software to improve efficiencies and revenue by analyzing over 4,000 specialized instructions and 5,000 configuration changes to determine their validity for removal or extension.
- Identified and prioritized which fault handling procedures should be automated, adhering to the goal of 70% increase in procedure automation by year-end, thus reducing the 7,500 daily customer service cases and reducing manpower.

Nexteer Automotive

Saginaw, MI

Software Engineering Intern | Python, MATLAB/Simulink, HTML

May 2023 – August 2023

- Partnered with senior software engineers to design and program an innovative software application which identifies requirement changes based on a Simulink model, reducing time to analyze by 95%.
- Transformed existing OEM requirements command-line-based system to a user-friendly interface.

Michigan State University

Midland, MI

Material Science Research Intern | Autodesk Fusion360

June 2022 – August 2022

- Analyzed and prioritized various biodegradable plastic materials using 3D printers, CAD, and nTopology to identify the ideal structures to facilitate wound healing and replace traditional bone fracture materials.
- Authored a formal research paper utilizing doctorate research methodology and presented to the American Chemistry Society.

PROJECT EXPERIENCE

Michigan Mars Rover (MRover) Project Team

Ann Arbor, MI

Teleoperations Team Lead | Vue.js, Django, Python, ROS, Linux, Git

August 2022 – May 2025

- Manage a team in designing and implementing a user-friendly interface to enable bidirectional communication with the rover.
- Accountable for team cohesiveness and sustainability including recruiting, onboarding, and knowledge transfer with the help of the MRover GitHub repository, dedicating on average 15 hours a week during the academic year.

University of Michigan

Ann Arbor, MI

SLAM Robotics | C++

August 2024 – December 2024

- Implemented LiDAR mapping, particle filter localization, and frontier-based exploration for autonomous robotic navigation.

NeRF Replication | Python, PyTorch

August 2024 – December 2024

- Implemented a neural radiance field (NeRF) model in PyTorch, successfully reproducing results from the CVPR 2020 paper.

FIRST Robotics

Midland, MI

Lead Programmer, Student Director | Java, Agile Methodology

September 2012 – May 2022

- Designed and troubleshooted autonomous robotic functionality and resolved escalations. Utilized SCRUM project management.

EXTRACURRICULAR ACTIVITIES

University of Michigan Marching Band, Member

August 2022 – December 2025